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A METEOROLOGY-DRIVEN LSTM FOR PREDICTING TIME SERIES OF VEGETATION LEAF AREA INDEX IN LAND SURFACE MODEL

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ABSTRACT

The leaf area index (LAI) is a basic vegetation structural variable in the ecosystem and climate model. Predicting vegetation LAI time series is essential for understanding the response of vegetation growth processes to climate change. Current predictive numerical phenology models are still subject to large uncertainties. Developing LAI time series prediction models is beneficial to improve the ability of land surface models to predict future vegetation growth. This study develops a deep learning model based on meteorological variables and attentional mechanisms to predict vegetation LAI time series. We tested our ALSTM for different biome types across North America and evaluated the model using satellite and flux tower data. The results demonstrated that the ALSTM outperforms other baseline models. The predicted ALSTM LAI shows high temporal and spatial consistency compared to satellite-based LAI. The ALSTM model can effectively predict downstream gross primary productivity (GPP) when coupled with the common land model (CoLM). ALSTM performs high accuracy in predicting the LAI of terrestrial vegetation and provides a reference for developing biophysical methods to predict future vegetation ecological processes for land surface models.

Index Term — LSTM, LAI, land surface model, deep learning, terrestrial vegetation

1. INTRODUCTION

Vegetation is the mainstay of terrestrial ecosystems, which plays an essential role in maintaining ecological balance and regulating the material cycle and energy flow between the various circles of the earth. The change in plant leaves is one of the most apparent activities in vegetation phenology, reflecting the impact of global climate and environmental changes on terrestrial ecosystems [1]. The amount of plant leaves is often quantified by the leaf area index (LAI), which is defined as the total plant leaf area per unit of land area. LAI is the intermediate or key variable that characterizes the canopy structure of vegetation in terrestrial ecosystem models, and it is involved in a series of physiological and ecological processes, such as radiative transfer, photosynthesis, respiration, and evapotranspiration [2]. Thus, the ability of the model to simulate and predict LAI largely determines the accuracy of the associated terrestrial ecosystem process model.

Ecosystems are complex systems with numerous fundamental processes that interact and constrain each other. Many factors affect ecosystem phenological processes, and their mechanisms are still not fully understood, therefor developing process-based numerical phenological prediction models is challenging [3]. Recent studies have emphasized the potential of deep learning methods in Earth system modeling, such as combining deep learning models with physical models to improve Earth system models and their components [4]. Machine learning methods provide a new means of modeling ecosystem processes, effectively complementing and improving current numerical physical models [5]. Many studies have built machine learning models to estimate LAI time series or critical phenological periods from observational data. Regression Neural Networks (GRNN) and BiLSTM combined with multi-temporal satellite reflectivity data have also been developed to produce continuous GLASS LAI products [6,7]. Dai et al. conducted a comparative study between machine learning and traditional phenology models, demonstrating that machine learning algorithms can improve the performance of phenology models. It is of interest to test the deep learning approaches on exploring the relationships between vegetation phenology and climate conditions, providing a reference for the developing biophysical models.

Recurrent neural networks have apparent advantages in time series processing and are widely used in the study of time series forecasting. The LSTM and its variant models (BiLSTM, GRU, etc.) dominate the field of time-series

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prediction due to their specific gating mechanism, allowing recurrent neural networks to make effective use of long-range time-series information [8]. We also note that some popular machine learning approaches, SVM, RF, and CNN, are also accelerating and assisting time series modelling and prediction in the field of remote sensing. The various machine learning methods have unique network structures and principles, and it is uncertain which scheme is most suitable for predicting vegetation growth processes based on meteorological variables. Therefore, it is necessary to evaluate the various prediction methods before carrying out prediction work on ecological processes. We also note that attention mechanisms have started to be adopted in the field of geographical forecasting. E.g., Tian et al. introduced an attention mechanism to improve the accuracy of crop yield estimation [9]. Compared to the above applications, research introducing attention mechanisms in predicting future vegetation growth trends is rare. Therefore, introducing attention mechanisms for LSTM in this study is expected to improve and enhance the performance of existing approaches for vegetation prediction.

The main goals of this study are to: (a) develop a meteorological data-driven and attention mechanism-based long short-term memory (ALSTM) model to predict terrestrial vegetation LAIl; (b) test the ALSTM model using satellite-derived LAI products and site data, and then input the CoLM2014 to compare predicted GPP differences to validate the performance of ALSTM in a land surface model.

2. STUDY MATERIALS

The Daymet meteorological dataset is a data product derived from modelling and extrapolation of ground-based observations, which provides accurate, long-term estimates of meteorological variables for North America. The Daymet meteorological variables include daily minimum and maximum temperatures, precipitation, vapor pressure, shortwave radiation, snow water equivalent, and day length on a 1 km x 1 km grid. We obtained the Daymet dataset from the Oak Ridge National Laboratory website (http://daymet.ornl.gov/) for the period 2001-2016. We used the driving data for a deep learning-based prediction of vegetation growth.

The satellite LAI products we used include 8-day composite MODIS LAI (MOD15A2H) and GLASS LAI (V5) at 1km resolution. MODIS LAI product is produced from the LUT (Look-up-Table) algorithm based on a three-dimensional radiative transfer model, which is closely aligned with site LAI and captures seasonal changes of LAI in different biomes. MODIS LAI provides accurate simulation and is widely used in land surface models such as CoLM [10]. Therefore, we define the MODIS LAI as the truth LAI in the process of deep learning modelling. GLASS LAI is based on a deep learning approach (GRNN) and surface reflectance, and Liang et al. has demonstrated the high accuracy of GLASS LAI in long time series, high-resolution LAI products with better spatio-temporal continuity, and smoother temporal profiles which require little pre-processing [6]. Thus, the GLASS LAI is used to compare and check the ALSTM LAI directly.

3. METHODS

Many deep learning methods are used for time series prediction, and to find the best deep learning model to predict LAI, four recurrent neural networks (RNN, LSTM, GRU, BiLSTM) commonly used for time series simulation and three common machine learning methods (CNN, SVM, RF) were selected for comparison.

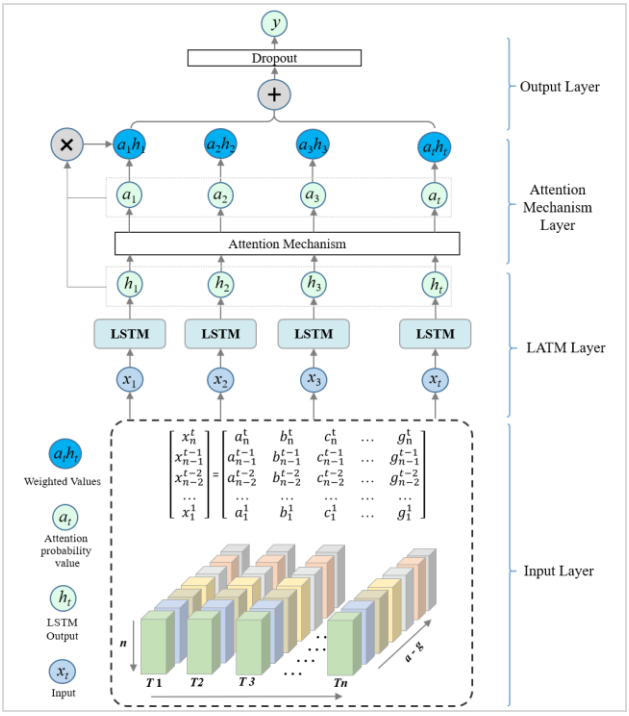


Fig 1 ALSTM model architecture

The attention mechanism is a resource allocation mechanism that mimics the attention of the human brain, allowing more attention to be allocated to features with high weights and relevance in the input time series, thus allowing the model to learn more key information from the input series data. The ALSTM architecture constructed in this subsection is derived from the test results in Section 4.1. To try to improve the LSTM model for predicting vegetation growth under future meteorological conditions, we introduced an attention mechanism on the original LSTM model and constructed the ALSTM model. The structure of the model is shown in Fig 1, where n is the number of features in the input, T denotes the length of the time series, and a-g denotes the meteorological variables in the input model.

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4. RESULTS

4.1 Model performance comparison

Table 1 compares the performance of different deep-learning algorithms for predicting LAI using meteorological variables. We divided the 5000 selected samples into three types: EV, DV, and SDV. Among the six baseline models (LSTM, GRU, BiLSTM, RNN, CNN, SVM), the LSTM model showed the best accuracy. Therefore, we chose the LSTM model and introduced an attention mechanism to improve the model, and the performance of the ALSTM model was improved.

Table 1 Evaluation of the different machine learning models

Model	EV			DV			SDV		
	R^2	RMSE	MAE	R^2	RMSE	MAE	R^2	RMSE	MAE
ALSTM	**0.86**	**0.556**	**0.382**	**0.93**	**0.503**	**0.354**	**0.90**	**0.252**	**0.145**
GRU	0.79	0.692	0.433	0.90	0.583	0.410	0.89	0.264	0.144
CNN	0.83	0.609	0.424	0.84	0.710	0.523	0.87	0.301	0.172
BiLSTM	0.85	0.584	0.396	0.88	0.640	0.421	0.85	0.332	0.162
RNN	0.85	0.580	0.388	0.81	0.823	0.482	0.89	0.27	0.148
LSTM	**0.85**	**0.590**	**0.399**	**0.91**	**0.545**	**0.391**	**0.88**	**0.280**	**0.153**
RF	0.84	0.619	0.414	0.89	0.578	0.436	0.87	0.305	0.177
SVM	0.79	0.684	0.449	0.87	0.637	0.457	0.86	0.311	0.176

4.2 Spatial consistency of ALSTM LAI

The distributions of LAI, MODIS LAI, and MAE in the four quarters are shown in Fig 2. Overall, the ALSTM model captured the spatial distribution of LAI relatively accurately in all four seasons, and both ALSTM LAI and MODIS LAI were within 0.6 m^2/m^2 of each other in most cases, showing a clear spatial consistency. This result also indicates that this deep learning model based on meteorological variables can effectively learn the seasonal distribution pattern of vegetation and shows a high similarity with observed LAI.

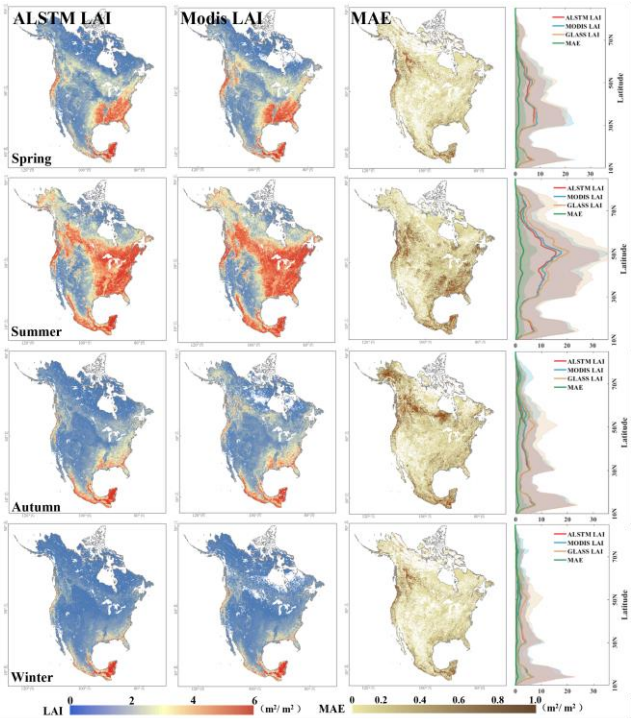


Fig 2 The spatial distributions ALSTM LAI, MODIS LAI, and MAE. This figure uses four typical days in spring (5 May), summer (1 August), autumn (5 November), and winter (6 February) in 2016.

4.3 Temporal consistency of ALSTM LAI

Fig 3 shows the time series comparison of ALSTM LAI, MODIS LAI, GLASS LAI, and FLUXNET LAI for 7 sites under different biome types from 2009-2016. ALSTM LAI has similar and continuous LAI time series divisions with MODIS LAI and GLASS LAI, which indicates that the ALSTM model can better learn the temporal change characteristics of vegetation growth.

We validate the validity of simulating future moment LAI time series based on meteorological variables and ALSTM at the site scale. The FLUXNET and Daymet meteorological variables entered into the ALSTM model yield consistent LAI time series under different vegetation types at different sites. These time series plots further illustrate the quality of the ALSTM LAI: it is a noise-free continuous time series that reflects seasonal changes in vegetation growth and is consistent with field measurements.

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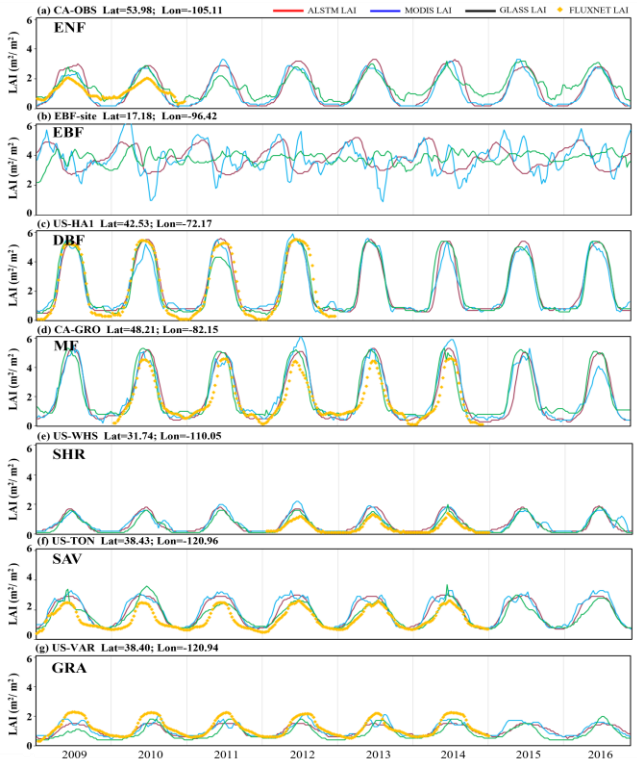


Fig 3 Comparison of ALSTM LAI, MODIS LAI, GLASS LAI, and FLUXNET LAI at FLUXNET sites for different biome types during 2009 - 2016.

4.4 Comparison of GPP calculated by CoLM

Fig 4 shows the spatial distribution of the two types of GPP results after the output from the CoLM calculation. The spatial distribution and the trend with latitude are highly consistent between the two types of GPP products, with only a few GPP pixels differing. In summer and autumn, the R^2 for both types of GPP products can reach above 0.95. The straight lines and shaded parts of Fig 4 shows the mean and standard deviation of the two types of GPP products at this latitude, which also almost overlap.

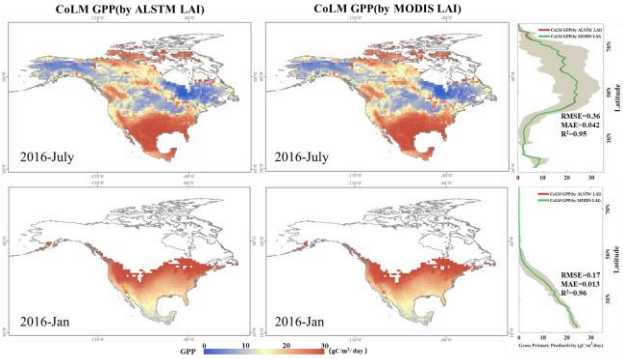


Fig 4 Comparison of GPP results obtained from the CoLM2014. Panel (a) and (c) are predicted GPP with ALSTM LAI. Panel (b) and (d) are predicted GPP with MODIS LAI. Panel (e) and (f) are latitudinal averages of GPP results.

The above experimental results show that the model-based prediction of ALSTM LAI and the satellite-based inversion MODIS LAI can obtain highly consistent GPP results after input to the CoLM model and that the assimilation of the ALSTM scheme to the land surface process model can effectively carry out the prediction of land surface variables.

5. CONCLUSIONS

ALSTM can effectively predict the leaf area index of terrestrial vegetation based on the relationship between future meteorological variables and vegetation activity, accurately capturing the spatial pattern and interannual variability of vegetation growth. Coupling the ALSTM model to the land surface model is an effective way to make predictions of land surface process variables. This study can be used as a reference for developing models for predicting future vegetation ecological processes in land surface models.

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